

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who has built the systems Arize Phoenix is designed to monitor - production RAG pipelines with ~35% precision improvement, agentic LLM workflows with >30% hallucination reduction, and predictive ML pipelines with 15-20% recall gains in healthcare. Hands-on with LangChain, LlamaIndex, FastAPI, and Docker. Passionate about translating production AI debugging experience into developer education, tutorials, and open-source demos for the LLM evaluation community.

CORE COMPETENCIES

RAG Systems • LLM Evaluation • Agentic Workflows • LangChain • LlamaIndex • Python • FastAPI • Docker • Embeddings • Vector Search • Prompt Engineering • Evaluation Pipelines • Open-Source Development

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

Kent, OH - Healthcare Technology, AI Consulting, SaaS Healthcare

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%. Retrieval-grounded response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall by 15-20% on high-risk categories while holding precision >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts and appointment histories; raised dataset reliability above 98% and cut downstream model instability by ~40%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana** to track long-running queries and CPU/I/O contention; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization system** across 5,000+ sessions; reduced manual review time by 60-70%. Achieved ~85% **alignment** between AI-selected highlights and human-curated moments.
- Implemented **RAG-based document Q&A** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with schema-validated JSON contracts between agents. RAG-grounded CPT/ICD validation; approximate-nearest-neighbor similarity search for duplicate detection.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, vector search

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Full-stack browser extension (Manifest V3) integrating four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/local). Multi-section capture pipeline with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 sites.

Python, TypeScript, JavaScript, Manifest V3, multi-provider AI APIs

Dream Decoder - AI-Powered Multimodal Creative Platform

Full-stack FastAPI + React/TypeScript/Vite app orchestrating Perplexity Sonar and Replicate image models. Intermediate structured prompt transformation layers improved contextual alignment ~30% and first-pass image success rate ~25-30%.

Python, FastAPI, TypeScript, React, Vite, Tailwind, multi-provider AI APIs

Agentic Pixel Character Synthesis & Animation Engine ongoing

Generative AI system for identity-consistent, pose-controlled pixel character synthesis. Stable Diffusion fine-tuning + LoRA, ControlNet + OpenPose/MediaPipe for pose conditioning, LLM-based agent orchestrator decomposing high-level prompts into generation tasks.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, LLM orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL), TypeScript, React

LLM & GenAI: RAG systems, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, embeddings, vector search, LangChain, LlamaIndex

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers

Computer Vision: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision